

SVKM's NMIMS University
Mukesh Patel School of Technology Management and Engineering

Program: B Tech/MBA Tech (Computer Engineering, Information Technology, Artificial Intelligence) B Tech (AIML, AIDS, Computer Science)				Semester: III/ IV /V/VI	
Course: Human Computer Interaction				Course Code: 702CO0E040	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks- 50)	Term End Examinations (TEE) (Marks- 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Prerequisite: Software Engineering					
Course Objective This course gives an introduction to Human Computer Interface and provides an understanding of user centered design process. It will help students to design and evaluate interactive systems keeping users in mind.					
Course Outcomes After completion of the course, students will be able to - 1. Discuss the importance of good interface design for human computer interaction 2. Apply design principles, models and evaluation techniques to user interface design 3. Identify various aspects of user experience and design thinking in HCI 4. Design user interface application using HCI concept					
Detailed Syllabus					
Unit	Description				Duration
1	Introduction Introduction to HCI, Importance of good interface design, Notions- Human, Computer & Interaction. Multi-disciplinary Applications of HCI.				04
2	Design Process & Interaction Introduction of design, Types of design: User-centered design, Participatory design, Scenario based design, Interaction design basics, users & persona, scenario				05
3	Design Rules Cognitive psychology - Visual perception, Ergonomics, Memory Models, Shneiderman's design rules, Norman's 7 principles for designing				06
4	HCI Models GOMS model, Hierarchical Task Analysis				03



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5	Evaluation Techniques in HCI Need of evaluation in interface designing, introduction to quantitative and qualitative research methods in designing, Types of evaluation techniques- Heuristics evaluation model, Experimental evaluation model.	06
6	User Experience Basic understanding of UX in HCI, Role of UI and UX in HCI designing, Elements of UX	04
7	Designing for Emerging Technologies Voice based UI, designing for wearables.	02
	Total	30

Text Books

1. Helen Sharp, Jennifer Preece, Yvonne Rogers *Interaction Design: Beyond Human-Computer Interaction*, 5th Edition, Wiley Publication, 2019.
2. Alan Dix, Janet Finlay, Gregory Abowd, Russel Beale, *Human-Computer Interaction*, 4th Edition, Pearson Education, 2009.

Reference Books

1. Ben Shneiderman, *Designing the User Interface: Strategies for Effective Human- Computer Interaction*, 3rd Edition, Pearson Education, 2014. ISBN-13: 9780134380384
2. Follett Jonathan (Ed), *Designing for Emerging Technologies*, 1st Edition, O'Reilly, 2014. ISBN:9789351109020
3. Levy Jaime, *UX Strategy: How to Devise Innovative Digital Products that People Want*, 1st Edition, O'Reilly, 2017.

Laboratory Work:

8 to 10 experiments (and a practicum where applicable) based on the syllabus.



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Program : B Tech/MBA Tech (IT, Computer Engineering), B Tech (CSE-Cybersecurity, CSBS, Computer Science, CSEDs, AIML, AIDS)				Semester : III,V	
Course : Software Engineering				Code : 702IT0C041	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks- 50)	Term End Examinations (TEE) (Marks- 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Prerequisite: Programming for Problem Solving					
Course Objective The objective of the course is to familiarize the students with Software engineering principles, practices and standards required to develop a quality software. The course also intends to develop the ability and skills for the task of requirement analysis, design and modelling.					
Course Outcomes After completion of the course, the student will be able to - 1. Explain the characteristics of various process models used in the development of a Software project 2. Demonstrate an understanding of various Analysis and Design models that provide a basis for the software development 3. Apply UML concepts for modeling software functionality for a given scenario 4. Create test cases for validating the working of the software developed					
Detailed Syllabus					
Unit	Description				Duration
1.	Importance of Software Engineering Role of Software, Categories of Software, Legacy Software, Software Myth.				03
2.	Prescriptive Process Models Process Framework, Capability Maturity Model Integration, Waterfall Model, Incremental & RAD Models, Prototyping, Spiral Model, Concurrent Development Model. Agile Process Models Agility, Agile Process, Extreme Programming, Adaptive Software Development, SCRUM				07
3.	UML Modeling Visual modeling with UML, Use case model, Modeling with classes, Identifying classes and objects of real world problems, Defining events and attributes, process of creating class diagram.				08




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	State diagram, Activity diagram, Modeling interaction and behaviour, Sequence and Collaboration Diagram.	
4.	Requirement Analysis & Design Requirement Engineering tasks, Elements of Analysis Model, Data Modeling Concepts, Data Flow Model, and Control Flow Model.	03
5.	Architectural Design Software Architecture, Data Design, Architectural Styles, Representing System in Context, Refining Architecture into Components, Mapping Data Flow into a Software Architecture.	03
6.	User Interface Design Golden Rules for User Interface Design, Interface Analysis & Design, Interface Design Steps.	02
7.	Testing Strategies & Software Quality Test Strategies for Software, Verification & Validation Testing, Unit Testing, Integration Testing, System Testing. McCall's Software Quality Factors, ISO 9126 Quality Factors, Process & Project Metrics, Metrics for Software Quality, SQA Activities, CMMI.	04
	Total	30
Text Books		
1. Pressman and Roger S., <i>Software engineering: a practitioner's approach</i> , 9 th Edition, McGraw Hill, 2019.		
Reference Books		
1. Sommerville and Ian., <i>Software engineering</i> , 10 th Edition, Pearson Education, 2017.		
Laboratory Work		
8 to 10 programming exercises (and a practicum) based on the syllabus.		




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Program: B Tech/MBA Tech IT				Semester: V	
Course: Embedded Systems				Code: 702IT0C039	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks- 50)	Term End Examinations (TEE) (Marks- 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Programming for Problem Solving, Operating Systems					
Course Objective The objective of this course is to acquire knowledge about the development of embedded system architecture using RTOS and enable the students to understand embedded-system programming and use it for designing and developing embedded solutions.					
Course Outcomes After completion of the course, the student will be able to - 1. Explain embedded system architecture and RTOS 2. Identify how microprocessor, memory, peripheral components and buses interact in an embedded system design 3. Develop solutions for various real time applications using Embedded programming					
Detailed Syllabus					
Unit	Description				Duration
1.	Introduction to Embedded Systems Classification of Embedded System, Concept of Embedded System Design, and Design challenges: Processor technology, IC technology, Design technology and Trade-offs.				04
2.	Architecture of Embedded System Hardware Architecture, Software Architecture, Embedded System product Development Life cycle.				04
3.	Real-Time Operating Systems (RTOS) RTOS Architecture, Selecting Architecture, Architecture of the kernel, Tasks and Task Scheduler, Scheduling algorithms, Interrupt Service Routines, Semaphores, Mutex, Mailboxes, Message queues, Event Registers, Pipes, Signals, Timers, Memory management, Priority Inversion problem.				06
4.	Processor and memory Organization Classification and brief overview of microcontrollers, microprocessors and DSP's. Introduction and Architecture of PAL, PLA, CPLD, FPGA, ASIC, PSOC.				06
5.	Overview of Hardware -Software co design Fundamental Issues in Hardware-Software co-design, Computation models in Embedded system design, Introduction to Unified Modeling Language (UML), Hardware Software Trade-offs.				04
6.	Embedded software development environments				06




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	Challenges and issues in embedded software development, Device drivers, System calls and Programming languages: assembly languages, high level languages like C/C++, Source Code Engineering tool for Embedded C/C++. Introduction to Embedded Java.	
	Total	30
Text Books 1. Raj Kamal, <i>Embedded Systems Architecture, Programming, and Design</i>, 3rd Edition, Tata McGraw Hill, 2019. 2. K.V. Shibu, <i>Introduction to Embedded Systems</i>, 2nd Edition, Tata McGraw, 2016.		
Reference Books 1. Valvano, <i>Embedded Microcomputer Systems: A real time interfacing</i>, 3rd Edition, Cengage Learning, 2013.		
Laboratory Work 8 to 10 programming exercises based on the syllabus.		




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Program: B Tech/MBA Tech IT /B Tech CSE-Cybersecurity				Semester: V	
Course: Visual Analytics				Code: 702IT0C040	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks - 50)	Term End Examinations (TEE) (Marks - 100)
0	2	0	1	Marks scaled to 50	-
Pre-requisite: Basic computer knowledge, Data Warehousing and Mining					
Course Objective The objective of the course is to familiarize the students with tools and processes to analyze datasets using visual representations of the data. The course helps students to visualize the data in graphs, charts, maps etc., and thereby identify patterns and develop actionable insights that help one to make better, data-driven decisions.					
Course Outcomes After completion of the course, student would be able to- 1. Demonstrate a working knowledge of data visualization 2. Create visualization charts appropriate for the underlying data 3. Apply modern statistical learning techniques on data visualizations 4. Design and implement interactive links between various objects in a data visualization report					
Detailed Syllabus					
Unit	Description				Duration
1.	Getting Started with Visual analytics What is data visualization, methodology for data visual analytics, Exploring Visual Analytics Concepts, Using Visual Analytics, Discussing the course environment and scenario.				2
2.	Data investigation and preparation Accessing data from local device/ server/social media. Data investigation to find common issues. Data preparation - cleaning, modifying the properties of data items, applying data source filter, creating new data items, joining tables/ columns, Data exploration with generated dataset.				3
3.	Using Visual analytics Explorer Examining the visual analytics explorations, selecting data and defining data items properties, creating visualizations, Enhancing visualization with advanced analytics.				5
4.	Data visual analytics Modern statistical learning techniques on data visualizations.				4
5.	Designing Reports with Visual Analytics Examining the Visual Analytics Designer, creating a simple report, working with graphs, working with filters and report sections, establishing interactions, working with gauges, working with tables, working with other objects.				8
6.	Viewing Visual Analytics Reports viewing reports on the Web, viewing reports on a mobile device				2
7.	Case study Create a visualization report and dashboard on a business scenario.				6




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	Total	30
Text Books		
1. Kieran Healy, <i>Data Visualization: A Practical Introduction</i> , 1 st Edition, Princeton University Press, 2018.		
2. Cole Nussbaumer Knaflic, <i>Storytelling with Data: A Data Visualization Guide for Business Professionals</i> , 1st Edition, Wiley; 2015.		
Reference books		
1. <i>Visual Analytics 7.5: User's Guide</i> by SAS Institute (online SAS documents).		
2. Trevor Hastie Robert Tibshirani Jerome Friedman, <i>The Elements of Statistical Learning: Data Mining, Inference, and Prediction</i> , 2 nd Edition, Springer, 2017.		
Laboratory Work		
8 to 10 Programming exercises (and a practicum) based on the syllabus.		



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Program: B Tech/MBA Tech IT, B Tech (AI and ML)				Semester: III / V	
Course: Management Information Systems				Code: 702IT0E040	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks- 50)	Term End Examinations (TEE) (Marks- 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Prerequisite: Database Management Systems					
Course Objective This course gives an introduction to Management Information Systems. It aims at explaining how Information Technology / Information Systems help organizations to achieve competitive advantage in the market.					
Course Outcomes After completion of the course, the student will be able to - 1. Explain the role of Information System in business environment 2. Analyze existing business applications in use and design a business application for a given scenario 3. Describe the implications of social media on organizations 4. Identify the challenges in the management of global information systems and assess the security and ethical concerns					
Detailed Syllabus					
Unit	Description				Duration
1.	Foundations of Information systems and Management Why Information systems? Fundamental behavioural, technical, business and managerial concepts about the components and roles of information systems, Types of information systems: Operations support systems and Management support systems, Definition, role and impact of MIS, MIS as a support to the management and a tool for management process.				03
2.	Strategic Use of Information Systems Fundamentals of strategic advantage –What is strategic advantage, brief overview of business strategy, organizational strategy and information systems strategy. Porter's five forces model.				04
3.	Using Information Technology for strategic advantage Major concepts, developments, and management issues in information technology (hardware, software, networks, data resource management, and other information processing technologies such as the Internet).				04
4.	Business Applications-I Major uses of information systems for the operations, management, and competitive advantage of an e-business enterprise. Electronic Business Systems – Enterprise business systems, Functional business systems Enterprise Resource Planning (ERP) systems – What is ERP? Benefits,				06




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	challenges and trends in ERP Customer relationship Management (CRM) systems - What is CRM? Three phases of CRM, Benefits, challenges and trends in CRM Supply Chain Management (SCM) systems - What is SCM? Role of SCM, Benefits, challenges and trends in SCM.	
5.	Business Applications-II Decision support systems (DSS) – Introduction, DSS components, Decision and management, information quality, Reporting, Online analytical processing, Using DSS – what-if analysis, sensitivity analysis, Goal-seeking analysis, Optimization analysis, Data mining for decision support. Knowledge Management and Business Analytics - Data, information and Knowledge, What is knowledge Management, Managing Knowledge to Business Intelligence, Why manage knowledge, Knowledge management processes, Components of business analytics.	05
6.	Social Media and Organizations Leveraging social media for business, What to track in social media, Network effects for organizations, Social computing applications (Enterprise social computing) and implications on organizations.	04
7.	Management Challenges Security, Ethical and Societal Challenges Global Management of Information Systems – Cultural, political and geo-economic challenges, Global business/IT strategies and applications, Global data access issues.	04
	Total	30
Textbooks 1. Kenneth C.Laudon and Jane P. Laudon, <i>Management Information Systems – Managing the Digital firm</i>, 14th Edition, Pearson Education, 2016. 2. James A O'Brien and George M Marakas, <i>Management Information Systems</i>, 11th Edition, Tata Mc-Graw-Hill, 2017.		
Reference Books 1. Effy Oz, <i>Management Information System</i>, 6th Edition, Cengage Learning, 2013.		
Laboratory Work 8 to 10 programming exercises based on the syllabus.		




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Program: B Tech IT/MBA Tech IT				Semester: V	
Course: Advanced Data Structures				Code: 702IT0E041	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks - 50)	Term End Examinations (TEE) (Marks - 100)
2	2	0	3	Marks Scaled to 50	Marks scaled to 50
Prerequisite: Programming for Problem Solving					
Course Objective The objective of this course is to familiarize students with advanced data structures used to solve real world problems.					
Course Outcomes After completion of the course, student would be able to- 1. Choose appropriate data structures and algorithms, understand the ADT/libraries, and use it to design algorithms for a specific problem. 2. Design and Implement solution for a given real time problem using hashing techniques 3. Use and Implement red-black trees, B-trees and Splay trees 4. Implement pattern matching algorithms for various text processing applications					
Detailed Syllabus					
Unit	Description				Duration
1	Dictionaries Definition, Dictionary Abstract Data Type, Implementation of Dictionaries.				3
2	Hashing Review of Hashing, Hash Function, Collision Resolution Techniques in Hashing, Separate Chaining, Open Addressing, Linear Probing, Quadratic Probing, Double Hashing, Rehashing, Extendible Hashing.				8
3	Trees Binary Search Trees, AVL Trees, Red Black Trees, 2-3 Trees, B-Trees, Splay Trees				5
4	Graphs Introduction, search operation, Topological sorting, Bellman Ford Algorithm and Dijkstra's Single Source Shortest Path Algorithm.				4
5	Text Processing String Operations, Brute-Force Pattern Matching, The Boyer-Moore Algorithm, The Knuth-Morris-Pratt Algorithm, Standard Tries, Compressed Tries, Suffix Tries, The Huffman Coding Algorithm, The Longest Common Subsequence Problem (LCS), Applying Dynamic Programming to the LCS Problem.				6
6	Design Techniques Divide and conquer, Greedy Algorithm; Dynamic Programming, Branch and Bound, Backtracking Techniques.				4




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	Total	30
Text Books 1. Thomas H. Cormen, Charles E. Leiserson, and R.L. Rivest. <i>Introduction to Algorithms</i>, 4th Edition, MIT Press, 2022 2. Narasimha Karumanchi, <i>Data Structures and Algorithms Made Easy: Data Structures and Algorithmic Puzzles</i>, 5th Edition, Careermonk Publications, 2016.		
Reference Books 1. Seymour Lipschutz, <i>Data structures with C</i>, 1st Edition, Schaum's Outlines, 2017. 2. M T Goodrich, Roberto Tamassia, <i>Algorithm Design and Applications</i>, John Wiley, 2015. 3. Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman. <i>The Design and Analysis of Computer Algorithms</i>, Pearson Education (Singapore) 2008.		
Laboratory Work 8 to 10 Programming exercises (and a practicum) based on the syllabus.		



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Program: B Tech/MBA Tech IT				Semester : V	
Course: Service Oriented Architecture				Code: 702IT0E042	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks- 50)	Term End Examinations (TEE) (Marks- 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Web Technology, Software engineering					
Course Objective					
To enable the students implement distributed web enterprise architectures using micro services.					
Course Outcomes					
After completion of the course, the student will be able to -					
1. Design and implement web services for creating enterprise web applications					
2. Explore semantic web to create ontologies and to query RDFs					
3. Execute web service integration, deployment and Security of micro services					
Detailed Syllabus					
Unit	Description				Duration
1	Distributed web Architecture Middleware, Messaging, Security and interoperability in web architectures. RPC and RMI for distributed web communication.				04
2	Service Oriented Architecture Introducing SOA- SOA triangle, layered architecture of SOA, Service messaging, service discovery, BPO - Business Process Outsourcing - Web service composition and coordination. Orchestration and Choreography.				07
3	Building SOA Web service creation and accessing -XML, JSON, WSDL, SOAP, UDDI, XML, Restful web services, BPEL.				06
4	Semantic Web: Mashup, Semantic Web Services, Metadata with RDFS Framework, Ontology, Owl, Logic for semantic web, SPARQL.				07
5	Micro services: Evolution, Modelling services, Integration, Deployment, Testing, Monitoring, Security. Implementation of micro services.				06
	Total				30
Text Books					
1. Allemang, and James Hendler., <i>Semantic Web for the Working Ontologist: Effective Modeling in RDFS and OWL</i> , 2 nd Edition, Morgan Kaufmann, 2011.					
2. Sam Newman, <i>Building Micro Services</i> , 2 nd Edition, O'Reilly, 2021.					
Reference Books					
1. Tarek Ziade, <i>Python Microservices Development</i> , 1 st Edition, O'Reilly publication, 2017.					




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| 2. Thomas Erl, <i>Service-Oriented Architecture: Concepts, Technology, and Design</i> , 1st Edition, Pearson Education, 2005. (Classic Book) |
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Laboratory Work

8 to 10 programming exercises based on the syllabus.
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Program: B Tech/MBA Tech IT				Semester : V	
Course: AI and Robotics				Code: 702IT0E043	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks- 50)	Term End Examinations (TEE) (Marks- 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Programming for Problem Solving					
Course Objective To enable students to build intelligent machines, software, or applications with a cutting-edge combination of machine learning and Robotics. This course discusses AI methods including search and Logical techniques, learning techniques and Robotics fundamental.					
Course Outcomes After completion of the course, the student will be able to - 1. Describe basic principles of AI toward problem solving, inference, perception, knowledge representation, and learning. 2. Explain and implement advance AI learning techniques 3. Apply knowledge of robotics, robot Kinematics, robotic programming and robot vision in developing solutions to real time problems					
Detailed Syllabus					
Unit	Description				Duration
1.	Introduction to AI History, Scope, agents, Environment, Automated Problem Solving, State Space Search, Heuristic Search, Game Tree Search.				04
2.	Logic and Deduction Propositional logic, Predicate Logic Fundamentals, Resolution Refutation, Constraint Satisfaction Problems.				05
3.	Learning Machine Learning Fundamentals, Learning Decision Trees, Neural Networks, Deep Learning Fundamentals, Reinforcement Learning Fundamentals.				06
4.	Introduction of Robotics Definition of a robot, types of robotic joints and motions, classifications of robot based on: Physical configurations, actuators and motion control; Terminologies used for robotics specification and selection for industrial applications.				04
5.	Robot Kinematics and Dynamics Homogeneous co-ordinates and co-ordinate transformations, kinematic parameters, use of Denavit-Hartenberg representation for finding arm equation of robotic arms, forward and inverse kinematics for basic industrial robotic configurations.				05
6.	Robot Vision and Programming				06




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	Sensing and digitization of vision data, image processing: image data reduction, segmentation, feature extraction, object recognition, and training of vision system, Robot programming methods, Robot Programming Languages.	
	Total	30
Text Books 1. Stuart Russell and Peter Norvig, <i>Artificial intelligence: A Modern Approach</i>, 4th Edition, Prentice Hall, 2020. 2. Saha S K, <i>Introduction to Robotics</i>, 2nd Edition , McGraw Hill Higher Education, New Delhi, 2014.		
Reference Books 1. M.P. Grover and N. G. Odrey, <i>Industrial Robotics: Technology, Programming and Applications</i>, 2nd Edition, TMH Edu. India, McGraw Hill India, 2017. 2. Kevin Warwick, <i>Artificial Intelligence: The Basics</i>, 1st Edition, Taylor and Francis, 2013.		
Laboratory Work 8 to 10 programming exercises based on the syllabus.		




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Program: B Tech/MBA Tech				Semester : V	
Course : Advanced Computer Networks				Code : 702IT0E044	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks- 50)	Term End Examinations (TEE) (Marks- 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Prerequisite: Computer Networks					
Course Objective					
The objective of this course is to familiarize students with the working of internetworks. This course focus on top-down approach, the Internet and a modern treatment of computer networking, its attention to both principles and practice, and its accessible style and approach toward learning about computer networking.					
Course Outcomes					
After completion of the course, the student will be able to -					
1. Describe medium access layer protocols					
2. Design IP addressing scheme, analyze packet forwarding, and compare different routing protocols					
3. Compare and analyze transport protocols and other key protocols essential for networking					
4. Explain various multimedia protocols, internet security concepts and future networks					
Detailed Syllabus					
Unit	Description				Duration
1.	Medium access layer Channel allocation problem, hidden node and exposed node problem FDM, TDM, CDMA, MACA.				03
2.	Internet protocol IPv4, compare IPv4 and IPv6, IPv6 - (address, special address, transition from IPv4 to IPv6; subnetting classless addressing).				05
3.	Network Layer Routing tables and forwarding, Routing protocols - Open Shortest Path First and Border Gateway Protocol, DHCP, Mobile IP, AODV.				06
4.	Transport layer Transmission Control Protocol - Reno, SCTP, I-TCP, Snoop TCP and Mobile TCP.				06
5.	Multimedia in the internet SIP, QoS (flow classes, flow control to improve QoS - FIFO, Priority and WFQ).				03
6.	Network security Introduction to Internet security - application layer security, transport layer security (SSL and TLS), IPSec, VPN, Firewall, WPA2, WPA3.				05
7.	Introduction to Internet of Things and applications, software defined networks and 5G.				02
	Total				30
Text Books					
1. Behrouz A. Forouzan, TCP/IP Protocol suite, 4 th Edition, McGraw Hill Education, 2017.					
2. James F. Kurose and Keith W. Ross, Computer Networking: A Top Down Approach, 7 th Edition, Pearson Education, 2017.					




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Reference Books

1. Behrouz A. Forouzan and Firouz Mosharraf, *Computer Networks: A Top-down Approach*, Special Indian Edition, McGraw-Hill, 2012.
2. Jochen Schiller, *Mobile communication*, 2nd Edition, Pearson Education, 2008.
3. Comer D., *Internetworking with TCP/IP Volume-I*, 6th Edition, PHI, 2015.
4. Jonathan Rodriguez, *Fundamentals of 5G Mobile Networks*, 1st Edition, Wiley, 2015.

Laboratory Work

8 to 10 programming exercises based on the syllabus.



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Program: B Tech and MBA Tech (All branches)				Semester: V	
Course: Design Experience II				Code: 702EE0C008	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks - 50)	Term End Examinations (TEE) (Marks - 100)
0	4	0	2	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: -					
Course Objective This course develops students' skills in human-centered design, interdisciplinary teamwork, and problem-solving through structured project work. Students will engage in stakeholder analysis, prototyping, testing, and iterative refinement while integrating engineering principles, creativity, and sustainability. Emphasizing collaboration and ethical responsibility, the course prepares students for industry and community-driven challenges by enhancing their project management, communication, and critical thinking abilities.					
Course Outcomes After completion of the course, the student will be able to - 1. Critically evaluate complex, real-world challenges using systematic problem-solving approaches. 2. Lead and collaborate in interdisciplinary teams to develop scalable and impactful solutions. 3. Apply advanced design methodologies and emerging technologies to prototype and refine innovative solutions in research. 4. Integrate sustainability, ethics, and feasibility analysis into the design and implementation process. 5. Demonstrate professional communication through technical reports, presentations, and stakeholder engagement.					
Detailed Syllabus					
Unit	Description				Duration
1.	The course covers Problem Analysis, where students assess the problem context, constraints, stakeholders, and solution stage while conducting literature reviews and benchmarking existing solutions. In Design Thinking & Concept Development, they apply the phases of Design Thinking – Empathize, Define, Ideate, Prototype, and Test – to develop and refine innovative solutions. Prototyping & Testing involves transforming ideas into practical prototypes, conducting experiments, usability testing, and iterative refinements. Through Project Management &				30

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	Documentation, students assign roles, track progress, and maintain structured records. Communication & Review focuses on preparing presentations, conducting design reviews, and incorporating feedback. Finally, Reflection & Improvement allows students to analyze challenges, enhance their skills, and document key learnings.	
	Total	30
Laboratory Work: The Laboratory work comprises various activities performed by the project teams for formulation, solution implementation and modification. Each team needs to submit the following documents: • Problem Statement Document • Project Development Report • Mid Semester Project Report • Research Paper / Conference Paper • Final Report		



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Program: All programs except CSBS and CSE(DS) 311(VT)				Semester: III/V	
Course : Game Design				Code: 702IT0I002	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks- 50)	Term End Examinations (TEE) (Marks- 100)
3	0	0	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: NA					
Course Objective This course aims to help course learners to produce a game's high concept document, one page blueprint, a physical prototype, pitch and supporting design documentation to move from a simple idea to a fleshed-out design, ready for implementation.					
Course Outcomes After completion of the course, the student will be able to - 1. Interpret the term "game," incorporating different concepts and multiple perspectives 2. Analyse the mechanics of paper and digital games, considering how these mechanics affect gameplay and player experiences 3. Design, develop, and test games using an iterative design process and considering user feedback 4. Create Design Documents using design template to communicate design ideas					
Detailed Syllabus					
Unit	Description				Duration
1.	Overview of Games Definition of Game, Genres of Games, Platforms of Game Distribution, Game Industry, Business of Game Publishing.				06
2.	Game Designing and Paper Prototyping Game Analysis Frameworks (Mechanics, Dynamics, Aesthetics (MDA) model and Design, Play and Experience (DPE) model), Layered Tetrad (Inscribed, Dynamic, Cultural layer), Iterative Design, Brainstorming and Ideation, Scoping, Design Goals, Math and Game Balance, Puzzle Design, Benefits of Paper Prototyping, Paper Prototyping Tools.				12
3.	Digital Prototyping Thinking in Digital Systems, Tools of Digital Prototyping, Introducing development environment Unity and development language C#, Variables and Components, Boolean Operations and Conditionals, Loops, Lists and Arrays, Functions and Parameters, Object Oriented Thinking, Agile Mentality.				09
4.	Making Design better User Experience Design, Playtesting, Functionality, Completeness and Balance, Fun and Accessibility.				09
5.	Communicating your Designs Visualization, Concept Art, Formats for Design Documents (Design Templates ex: One Page Design Template, High Concept Template etc.), Selling your ideas to the Game Industry].				09



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	Total	45
Text Books 1. Jeremy Gibson, <i>Introduction to Game Design, Prototyping, and Development</i>, Pearson Education, (ISBN-13: 978-0321933164, ISBN-10: 0321933168), 2015. 2. Tracy Fullerton, <i>Game Design Workshop-A Playcentric Approach to Creating Innovative Games</i>, 4th Edition, CRC Press Taylor and Francis Group, (ISBN-13: 978-1138090770), 2018.		
Reference Books 1. Ernest Adams, <i>Fundamentals of Game Design</i>, 3rd Edition, Pearson Education, (ISBN-13: 978-0321929679, ISBN-10: 0321929675), 2013. 2. Jesse Schell, <i>The Art of Game Design: A Book of Lenses</i>, 2nd Edition, CRC Press Taylor and Francis Group, (ISBN-13: 978-1138413696, ISBN-10: 1138413690), 2008.		
Reference Papers and eCourse 1. Winn, B., <i>The Design, Play, and Experience Framework. Handbook of Research on Effective Electronic Gaming in Education</i>, Richard Ferdig (editor). Volume 3, Chapter 58., 2009. 2. Hunicke, R, LeBlanc M., and Zubek, R., <i>MDA: A Formal Approach to Game Design and Game Research</i>, 2004. (Classic Book) 3. Reference eCourse: <i>Coursera: Principles of Game Design</i>, Michigan State University.		



Signature
(Head of the Department)

